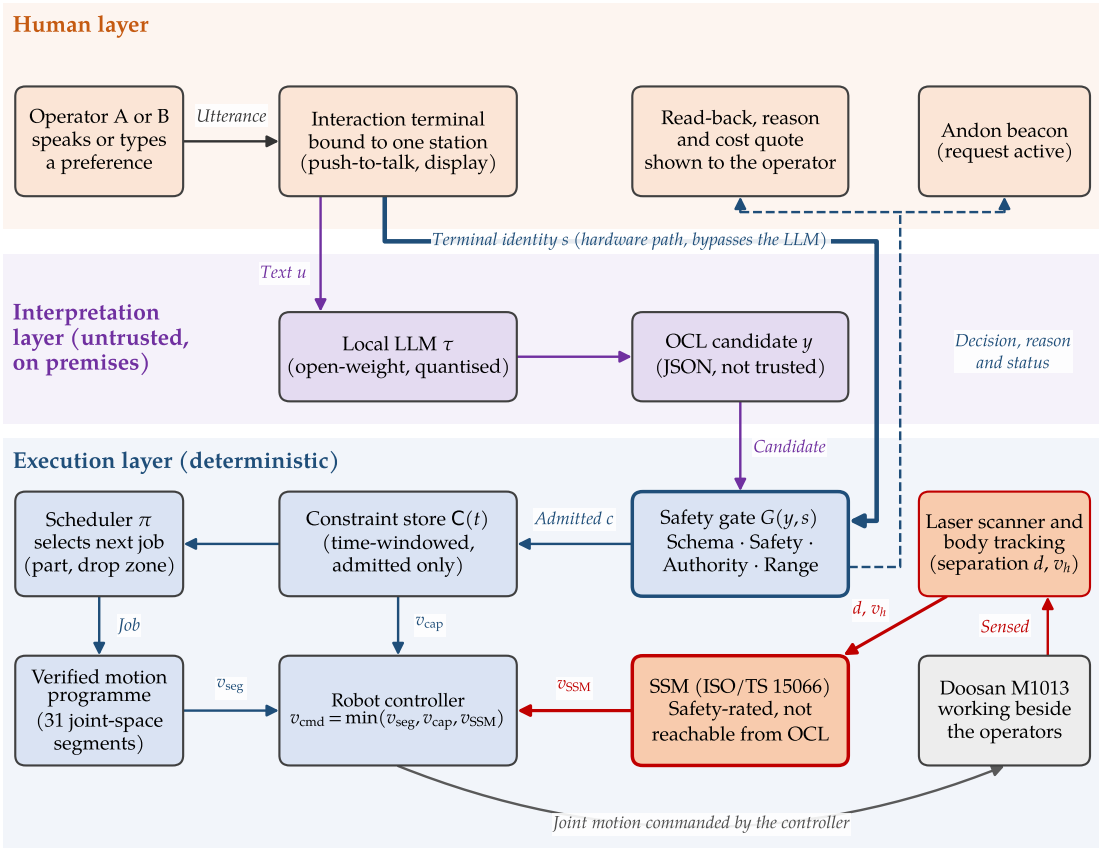
Human layer
Operator A or B speaks or types a preference
Utterance
Interaction terminal bound to one station (push-to-talk, display)
Read-back, reason and cost quote shown to the operator
Andon beacon (request active)
Terminal identity s (hardware path, bypasses the LLM)
Text u
Interpretation layer (untrusted, on premises)
Local LLM τ (open-weight, quantised)
OCL candidate y (JSON, not trusted)
Decision, reason and status
Candidate
Execution layer (deterministic)
Scheduler π selects next job (part, drop zone)
Constraint store C(t) (time-windowed, admitted only)
Admitted c
Safety gate G(y, s) Schema · Safety · Authority · Range
Laser scanner and body tracking (separation d, v_h)
d, v_h
Sensed
Job
v_cap
Verified motion programme (31 joint-space segments)
v_seg
Robot controller v_cmd = min(v_seg, v_cap, v_SSM)
v_SSM
SSM (ISO/TS 15066) Safety-rated, not reachable from OCL
Doosan M1013 working beside the operators
Joint motion commanded by the controller